

4/01/25
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ENVIRONMENT EFFECTS

Syllabus.

① Pollution & Air pollution. $\rightarrow$ e-waste

$\rightarrow$ Definition of pollution & pollutant, causes of pollution, Types of pollution - Air & Water pollution.

① Pollution :-

It may be defined as addition of unwanted foreign matters into the environment thereby changing the natural quality of environment and causing damage to human, human, plant and animal life.

② Pollutant :-

Different particulates such as dust, smoke, clay, ores, lead which pollutes the atmosphere and water are called pollutant.

③ Causes of pollution :-

- Cutting of trees
- radioactivity
- Volcanic eruption
- Strong wind
- Forest fire
- Sand
- minerals
- rapid industrialisation

* Types of Pollution :-

- Air ~~pot~~ pollution
 - Water pollution
- source & effect

* Air pollution :- It may be defined as the presence of one or more substance combined of one or in air may be injurious to humans with the comfortable enjoyment of life or properties.

* Types of Air pollution :-

→ On the basis of physical state there are four major group of air pollutant.

- Gases like SO_2 , SO_3 , CO , NO , CO_2 etc.
- Particulants - ~~the~~ dust, Smoke, asbestos.
- lead, dusts, smog.
- deforestation
- Radioactivity gases
- I.C engine

* Global warming or Green house effect.

→ The increasing of temperature of atmosphere is called as global warming or green house effect.

The gases responsible for global warming are CO_2 , CH_4 , CFCs , O_3 , NO_2 , water, vapour.

These gases absorb more light from the sunlight. Hence the temperature increase. If the global warming temperature increases by 1°C , the following effects are predicted.

- (i) The ice caps of polar regions melt thereby increasing the sea water level. Hence low lying areas are submerged in the sea water.
- (ii) The rate of evaporation of water from the hydrosphere increases which leads to cyclones, seasonal rains etc.
- (iii) The agricultural production decreases due to the fast evaporation of surface.

* Precautions: -

The global warming can be prevented or reduced by plantation.

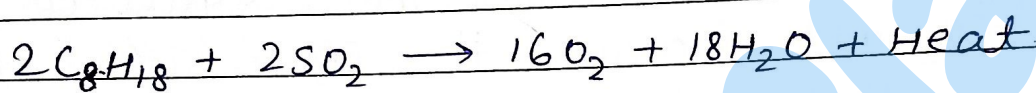
• Effects of air pollution on human beings: -

- (i) Carbon monoxide - Blood poisoning leads to death.
- (ii) Sulphur dioxide - Suffocation, aggravation of asthma and bronchitis irritations of throat and eyes.
- (iii) Hydrogen Sulphide: - Headache, Conjunctivitis

respiratory trouble, blockage of oxygen, asphyxiation and death.

* Air pollution due to I.C engines :-

→ I.C engines used in cars, buses, trucks etc. is one of the major air pollution. These engines use fuels hydrocarbons, Octane (C_8H_{18}) is a chief component of gasoline. When a mixture of gasoline and air burns in I.C engine, the reaction is



I.C engine emissions mainly consist of a mixture of CO , NO_2 etc. which are dangerous to all human beings.

* Radioactivity gases :-

→ Radioactivity elements which occur in rocks and soil as derivatives of uranium, thorium and actinium emit radioactive gases in air.

These gases are highly harmful to human health as they affect the chromosome and genes. They cause cancer effect of these gases can be seen for several years.

* Deforestation :-

→ Plants maintain the balance of CO_2 and O_2 in

nature because plants take up CO_2 for their photosynthesis and thus purify air.

• Definition :-

→ Deforestation means cutting of plants, trees and forest by man which disturbed the balance CO_2 and O_2 in the nature.

Disadvantage :-

i) Deforestation decreases the overall concentration of CO_2 and increases the concentration of CO_2 in the nature.

(ii) It will also affect on the control of hydrogen Sulphide, nitric acid and Cl . All this will lead to air pollution.

* What is the Causes of air pollution?

→ The important particulates which causes air pollutions are :-

i) Dust

(ii) Smoke

(iii) Smog

(iv) Asbestos dust

(v) lead dust.

(1) Dust :- Main source of dust are mines and

factories furnace ashes. Powerhouse Vehicular traffic, house cleaning dust, agricultural forest fire natural winds, combustion operation earth moving engine exhausts etc.

• Effects :-

In human being atmospheric dust causes silicosis if ~~dust~~ dust contains silica on property dust causes corrosion.

(2) Smoke :- It is composed of tiny particles of carbon, ash, oil, etc.

It is formed by incomplete combustion of fuel. The major sources of smoke emission are Rail, roads, locomotives, domestic wood, coal grates, Industrial power plant, open firest diesel engines, gasoline engines heaters etc.

• Effects :- (i) Loss of fuel value through imperfect combustions.

(ii) Spoiling of clothing, rags, ~~exterior~~ exterior finish of buildings.

(iii) The increase in cancer rate is related with the Smoke.

(3) Smog :- It is a combination of smoke and fog. fog means minute liquid that remain suspended in air.

• Effects :- It has same effect as smoke but some what prolonged one.

1) Lack of visibility of road causing accidents.

(i) Irritation eyes

(ii) Lung disorder

(iv) Damage of plant life

(4) Asbestos dust :- Its means sources are mining, processing and manufacture of asbestos gaskets and ropes used in automobiles asbestos building flooring and insulating materials.

Effects :-

(i) In human beings - It causes all types of respiratory diseases it also causes Asbestosis diseases.

(ii) On property It causes corrosion and soiling.

(5) Lead dust :-

→ It arises from lead mining and smelting works automobiles exhausts lead batteries, lead paints and manufacture of lead based alloy.

Effect :- (i) The lead dust settles down on plants and food stuff intended for human and animal consumption and causes lead poisoning.

* Explain the role of SO_2 , SO_3 , CO , NO_2 , CO_2 gases and their effects as air pollution.

OR

CO is a colourless and odourless gas and even then it is a pollutant why?

→ Sulphur dioxide (SO_2) :-

SO_2 is released from thermal power plants in which sulphur containing coal and diesel are fixed. Other sources are petroleum industry, oil refinery, sulphuric acid plants etc.

Effects :- The presence of this gas causes heart-respiratory diseases to man much as asthma, bronchitis, bronchitis, damages to agriculture, eye irritation, throat troubles, corrosion of metals. Some effect of SO_2 in atmosphere. On plants are cell membrane damage, chlorophyll destruction, growth and yield reduction etc.

(ii) Sulphur trioxide (SO_3)

→ It is formed by the oxidation of sulphur dioxide under the influence of sunlight.

Effects :- Even 1 ppm of SO_3 in air causes severe breathing discomfort and is irritating to the respiratory tract. SO_3 is converted into sulphuric acid in the atmosphere in the presence of water vapours which is responsible for acid rain. Acid rain has bad effect on material like cloth, papers, leather, buildings.

(iii) Carbon monoxide (CO) :- It is produced by the partial combustion of fuel in automobile, industries and oil refineries, cigarette, smoke and

domestic heat & appliances are the both source of CO. Though it is colourless, odourless, non-corrosive, yet it is very toxic in nature.

Effects : -

- (i) It reacts with the haemoglobin of blood and form the stable complex compound carboxy haemoglobin which reduces oxygen carrying capacity of blood to the tissues.
- (ii) It ultimately causes headache visual difficulty paralysis.
- (iii) It is toxic in nature and affects cardio vascular system thereby causing heart diseases.
- (iv) NO (Nitrogen oxide) & NO_2 : - It is obtained from the combustion of fuels (coal, diesel, petrol etc) Other sources are acid manufacture, explosive industry and acid-pickling plants and fertilisers industries.

Effect :- It causes respiratory illness among children. It is unpleasant in odour and causes irritation of eyes and even lungs congestion. It leads to the formation of petro-chemical smog. In sunlight nitrogen dioxide and hydrocarbons form smog from the combustion of petroleum products is called petrochemical smog which causes eye irritation, difficulty in breathing, thereby lungs asthma, bronchitis may result in non living around it.

(v) Carbon dioxide :- Carbon dioxide is produced by burning of fuels such as coal, wood, petroleum, products and gaseous fuels. It is also released into air by the respiration of plant and animals.

Effects :- The excess of CO_2 in the atmosphere also causes respiratory disorder and suffocation. It has been estimated that with 10 to 15% increase in CO_2 in the atmosphere, the world wide atmospheric temperature has been raised by 0.5°C .

* Write the causes of air pollution due to vehicle?

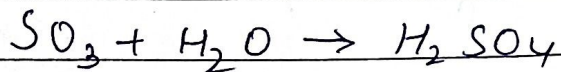
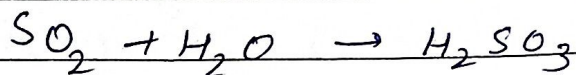
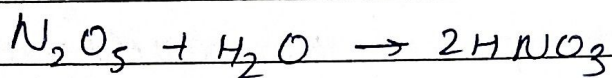
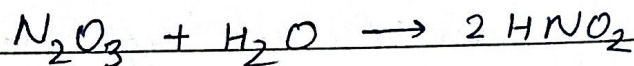
→ Vehicles are cause of major air pollution made by man. The internal combustion engine which is used for running motor car, buses, trucks, Vans, Scooters etc is major air pollutant.

When a mixture of gasoline vapour and air burns in I.C engine of an auto vehicle following reaction take place.



Carbon enters the atmosphere Besides this CO , unburnt C , SO_2 , hydrocarbon such as CH_4 acids, H_2O etc. enter the atmosphere. Thus when fuels like Kerosene, diesel petrol is burnt a lot of smoke is released into the atmosphere.

* Acid Rain :- The rainwater containing more amount of acid is called acid rain. It is due to oxides of nitrogen & sulphur. The nitrogen & sulphur oxides dissolved in the rain water & form nitrous acid, nitric acid, sulphurous acid & sulphuric acid.



* Effects of Air pollution. (Acid rain)

- (1) It causes deterioration of building especially made of natural marble.
- (2) It damage metals & can furnished.
- (3) It damage stone statues.
- (4) The life of aquatic animal is effected.
- (5) The fertility of the soil is decreased which results in low agricultural production.
- (6) The acidity of drinking water increase which causes activity in human beings.

* Depletion of Ozone layers: -

→ The Ozone layers present in the atmosphere acts like protective layer for U.V. light i.e. it prevents entering of U.V. light into atmosphere. The depletion of Ozone layer is due to the chloro fluoro carbons nitric oxide (NO) and Chlorine.

* Chlorofluoro carbons or freons: -

→ Chlorofluoro carbon are liberated from air conditioner and industries NO₂ is liberated from supersonic jet planes and chloronic is liberated due to eruption of dove. They are stable in atmosphere due to high temperature. They react with Ozone and causes holes in the Ozone and causes hole in the ozone layer. Hence, the following effect are observed.

- (i) Skin disease
- (ii) Eye disease like cataract formation
- (iii) Increasing of atmospheric temperature.
- (iv) Life of aquatic animals is affected.
- (v) Decrease photosynthesis in plants.

It should be controlled by reduction using C.F.S.P.

6/01/25 uo (2) WATER POLLUTION

(1) Definition of Water pollution?

→ Any alteration in the physical chemical and biological properties of water as well as contamination with any foreign substance, which would constitute a health hazard or otherwise decrease the utility of water is called water pollution.

(2) Causes of water pollution: -

- i) Domestic sewage
- ii) Industrial waste
- iii) Suspended particles
- iv) oily water
- v) Radioactive material
- vi) Heat

(3) Effects of water pollution: -

→ i) Bacteria causes cholera dysentery typhoid etc.

ii) Parasite worms cause round-worm and hook-worm infection.

iii) Viruses cause enteroviral infections.

iv) Algae causes foul-odour turbidity.

(V) Suspended matter in water makes murky.

(VI) Bicarbonate, chloride and sulphate of calcium and magnesium. Cause alkalinizing laxative effect. hardness and Corrosive effect on boilers leads causes less of appetite constipation, mental retardation.

★ How water pollution is prevented?

→ The water pollution can be prevented and controlled by the recycling of wastes e.g. - conversion of coconut and agricultural wastes into paper and board and Orbiarz gas plant provides not only gas but also inviched manure.

* Methods of preventing water pollution :-

- (i) Reutilisation of waste water
- (ii) Recycling of the waste.
- (iii) Stabilisation of the ecosystem
- (iv) Treatment of waste water.
- (v) Oxidation method
- (vi) Removal of pollutants
- (vii) plantation / forest management
- (viii) pollution regulating rules
- (ix) Dilution
- (x) using solar power.

* Domestic waste :-

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→ The domestic or sewage include human and household waste water, municipal waste water etc, directly drain into rivers cause the pollution of major rivers of our country.

• Effects :-

→ The Sewage contains human excrete, urine, kitchen waste, street waste and organic substances that provide nutrition for bacteria. Due to this, the spread of water borne diseases like dysentery, cholera, typhoid, hookworm infection, malaria etc takes place.

* Industrial waste :- Water get polluted by acids, alkalis, detergents, soaps, copper, zinc, lead mercury, insecticides etc. which are released from chemical industries. Waste from fertilizer plant contain nitrates phosphate ammonia all cause water pollution.

→ Effects :- Water containing toxic substance damage the biological activity and kills useful organisms.

② BOD :-

→ Biological Oxygen Demand (BOD) of Sewage is defined as the amount of free oxygen required for the biological oxidation of the organic matter under aerobic condition of 20°C & for a period of

five always.

* Determination of BOD :-

- (i) This test consist of measurement of dissolved oxygen content of the sewage before & after in incubation at 20°C for 5 day.
- (ii) If the last sample does contain any oxygen then it is diluted with water containing nutrients for bacterial growth whose ~~dimensional~~ dissolved oxygen content is predetermined.
- (iii) The whole solution is incubated in a closed bottle at 20°C for 5 days. After this oxygen is determined.

* Application of BOD :-

- (i) Larger the ~~concentration~~ of concentration of decomposable organic matter greater is the BOD and consequently more strength.
- (ii) BOD is considered as the major characteristics used in stream pollution control.
- (iii) It enable us to determine the degree of pollution at any time in the sewage.

* COD :-

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→ Chemical oxygen demand (COD) is an amount of oxygen required for the chemical oxidation of organic matter in sewage. It is a measure of oxidisable impurities present in the sewage.

This is an important and quickly measured parameter for sewage & industrial waste to determine their pollution strength.

* Determination of COD :-

→ (i) The organic matter of sample is oxidised to matter CO_2 & ammonia.

(ii) It is measured in ppm.

(iii) The oxygen is equivalent of $\text{K}_2\text{Cr}_2\text{O}_7$ consumed is taken as measure of COD.

(iv) In COD Test, both biologically oxidisable and biologically inert matter are oxidised.

(v) The COD value for a sample is always greater than BOD value.

* Application of COD :-

(i) COD determination is faster & preferable.

(ii) COD account for the total organic & inorganic matter oxidisable in waste water.

(iii) The Organic matter consist of C, H, S which on oxidation forms CO_2, H_2O, NO_2, SO_2 etc for this oxygen is required. It is through dissolved oxygen in effluent.

(iv) Hence it is essential to check the COD of effluent water before leaving it out for industry.

Difference b/w BOD and COD.

→ BOD	COD
(1) It requires 5 day long time.	(1) It require 2 to 5 hour.
(2) Numerical value of BOD for a waste is smaller.	(2) Numerical value of COD for same water is higher.
(3) Cellulose is not attacked in BOD test.	(3) Cellulose is attacked in this test.
(4) BOD test is influenced by PH toxic material types of microorganism.	(4) COD test is not influenced as such.
(5) It is used to measure degree of water pollution & waste strength.	(5) It is used to measure strength of organic waste matter.

* Biomedical Waste :-

→ The waste that contains human pathogens of sufficient virulence and unsufficient concentration that exposure to them susceptible host could result in disease is called as biomedical waste.

Effects of Biomedical wastes :-

→ (i) Waste generated by veterinary colleges, hospitals, animal houses can cause serious diseases like tuberculosis.

→ (ii) If plastic based biomedical wastes are incorporated scientifically the society has to face risk of being exposed to toxins when they are coming quick.

* Control Measures for biomedical waste :-

→ use of incinerator

→ use of safety box

→ encapsulation technique

→ Autoclaving or steam sterilization.